

Terraform-Automated-Linux-Bootstrap-on-Azure

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Project Overview

This project provisions a fully automated Linux VM environment on Microsoft Azure using Terraform and cloud-init.

The VM automatically configures:

- Nginx
- Docker
- Azure Monitor Agent
- Secure SSH
- Health check scripts
- Startup automation
- Custom HTML deployment
- Logging

Architecture

Terraform → Azure VM → cloud-init → Install & Configure Services

Technologies Used

- Terraform
- Microsoft Azure
- Ubuntu Linux

- cloud-init
- Docker
- Nginx
- Azure Monitor Agent

Project Structure

terraform-automated-linux-bootstrap-on-azure/

■

■ ■ ■ main.tf

■ ■ ■ provider.tf

■ ■ ■ variables.tf

■ ■ ■ terraform.tfvars

■ ■ ■ outputs.tf

■ ■ ■ network.tf

■ ■ ■ security.tf

■ ■ ■ vm.tf

■ ■ ■ monitor.tf

■

■ ■ ■ cloud-init/

■ ■ ■ ■ cloud-init.tpl

■ ■ ■ ■ custom-index.html

■ ■ ■ ■ health-check.sh

■

■ ■ ■ scripts/

■ ■ ■ ■ startup.sh

■

■ ■ ■ README.md

Prerequisites

- Azure Subscription
- Terraform Installed
- Azure CLI Installed
- SSH Key Pair

Setup Instructions

1. Clone Repository

```
git clone <repo-url>  
cd terraform-automated-linux-bootstrap-on-azure
```

2. Login to Azure

```
az login
```

3. Initialize Terraform

```
terraform init
```

4. Validate Configuration

```
terraform validate
```

5. Plan Infrastructure

```
terraform plan
```

6. Deploy Infrastructure

```
terraform apply -auto-approve
```

SSH Access

```
ssh -i ~/.ssh/bootstrap_key azureuser@PUBLIC_IP
```

Verify Services

Verify Nginx

```
systemctl status nginx
```

Verify Docker

```
systemctl status docker
```

Verify Health Check

```
bash /usr/local/bin/health-check.sh
```

Expected Output:

```
{ "nginx_status": "active" }
```

Verify Website

Open browser:

`http://PUBLIC_IP`

Expected:

Linux Bootstrap Completed Successfully

Logging

Startup Logs

```
cat /var/log/startup.log
```

Bootstrap Logs

```
cat /var/log/custom/bootstrap.log
```

Security Features

- SSH Key Authentication
- Disabled Root Login
- Disabled Password Authentication
- NSG Rules for Controlled Access

Common Issues

Nginx Service Missing

Cause:

Broken cloud-init YAML indentation.

Fix:

Use Terraform indent() function inside vm.tf.

Destroy Infrastructure

terraform destroy -auto-approve

Key Learning Outcomes

- Infrastructure as Code
- Azure VM Automation
- cloud-init Bootstrapping
- Terraform templatefile()
- Linux Service Management
- SSH Security
- DevOps Troubleshooting

Author

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